

■ DantShaar AI

Autonomous Dental Intelligence Assistant

Fine-Tuning (LoRA/QLoRA) + Hybrid Graph RAG Architecture

Technical Architecture & Product Documentation — v1.0 | 2026

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01 — Executive Summary

What DantShaaf Is and Why It Matters

DantShaaf is an autonomous AI dental assistant built for the 2026 agentic era. It is not a diagnostic app — it is a conversational agent that analyzes teeth conditions via computer vision, remembers patient history, proactively sets reminders, recommends products through a Graph RAG pipeline, and connects users with the nearest highest-rated dentist — all through natural conversation in Urdu and English.

The core technical innovation lies in a two-engine architecture: a Fine-Tuned Vision-Language Model (using LoRA/QLoRA on dental datasets) for accurate teeth condition recognition, combined with a Hybrid RAG system (dense + sparse + graph retrieval) for intelligent product and dentist recommendations. This combination delivers clinical-grade diagnosis accuracy with personalized, context-aware recommendations.

Metric	Target Value	Basis
Teeth Condition Recognition Accuracy	≥ 87%	Fine-tuned LoRA model on dental dataset
Product Suggestion Relevance Score	≥ 0.82 NDCG	Hybrid RAG with Graph traversal
Dentist Match Precision (top-3)	≥ 91%	Geo-filter + rating + specialty graph
Avg. Response Latency	< 2.5 seconds	Quantized QLoRA inference + vector cache
Autonomous Follow-up Rate	100%	MCP alarm + WhatsApp agent tools

Table 1 — Key Performance Targets

02 — Product Vision & Problem Statement

Who We Serve and What We Solve

The Problem

- 80% of dental problems in Pakistan/UAE go undiagnosed until they become severe due to cost, fear, or lack of awareness.
- Existing dental apps only diagnose — they do not act. They show a result and leave the user alone.
- Users forget follow-ups. No system autonomously tracks progress or reminds the patient.
- Dentist discovery is fragmented — Google Maps has ratings but no specialty-problem matching.
- Dental product recommendations are generic, not tied to individual diagnosis or history.

The DantShaat Solution

Feature	What It Does
Conversational Agent	User talks naturally — in Urdu or English. Agent handles everything else.
Vision-Based Diagnosis	Fine-tuned LoRA model analyzes teeth photos with clinical accuracy.
Autonomous Follow-up	MCP tools set alarms, send WhatsApp reminders, track progress — zero user effort.
Graph RAG Products	Product suggestions are tied to user's specific diagnosis + history + severity.
Dentist Matching	Nearest highest-rated dentist filtered by specialty matching user's exact problem.

Table 2 — Solution Feature Matrix



DantShaat's core differentiation: Every other dental app stops at diagnosis. DantShaat acts — it books appointments, sets alarms, orders products, and follows up. This is the agentic shift of 2026.

03 — System Architecture Overview

High-Level 5-Layer Design

DantShaaf is structured as five interconnected layers. Each layer has a single responsibility and communicates via defined interfaces. The architecture is designed for independent scaling — the inference layer (fine-tuned model) can scale separately from the retrieval layer (Hybrid RAG).

Layer	Name	Components	Responsibility
L1	User Interface	React Native App, WhatsApp Bot	Conversation, photo capture, progress view
L2	Agent Core	Claude Sonnet 4 + MCP Server	Orchestration, tool calling, memory, persona
L3	AI Inference	Fine-tuned LoRA/QLoRA Vision-LM	Teeth image analysis, condition classification
L4	Retrieval Engine	Hybrid RAG: Dense + Sparse + Graph	Product suggestion, dentist matching, history
L5	Data & Storage	PostgreSQL, Qdrant, Neo4j	Patient data, vector store, knowledge graph

Table 3 — Five-Layer Architecture

Inter-Layer Communication Flow



User sends a message or photo → Agent Core (Claude) decides which tools to call → Tool Router dispatches to Fine-Tuned Model (for images) or Hybrid RAG (for product/dentist queries) → Results are merged by the agent → Final response is generated and actions are executed autonomously.

04 — Fine-Tuning Pipeline: LoRA & QLoRA

Training a Dental-Specialist Vision Model

The base model cannot reliably distinguish dental conditions from standard photos. Fine-tuning on a curated dental dataset using LoRA (Low-Rank Adaptation) and QLoRA (Quantized LoRA) produces a lightweight, fast, accurate model that understands dental anatomy — without requiring a full model retrain.

Why LoRA and QLoRA — Not Full Fine-Tuning

Approach	Trainable Params	VRAM Required	Accuracy	Cost	Use in DantShaaf
Full Fine-Tune	100%	80GB+	Highest	Very High	Not feasible
LoRA	0.1–1%	16–24GB	Near-full	Low	Production model
QLoRA (4-bit)	0.1–1%	8–12GB	~95% of full	Very Low	Edge / mobile inference
Prompt Engineering	0%	Minimal	Limited	None	Fallback only

Table 4 — Fine-Tuning Method Comparison

Dataset Strategy

Dataset Source	Type	Approx. Size	Labels
UFBA Dental X-ray Dataset	X-Ray Images	4,000+	Cavity, Filling, Crown, Decay level
Tufts Dental Database	Radiographs	1,000+	Tooth numbering, Pathology
Kaggle Dental Condition Sets	Clinical Photos	8,000+	Gingivitis, Plaque, Healthy, Discoloration
DantShaaf Collected (Phase 2)	User Photos	Growing	User-verified + Dentist-confirmed labels
Synthetic Augmentation	Generated	5,000+	Rotations, lighting, occlusion variants

Table 5 — Training Dataset Sources

Base Model Selection

Model	Type	Params	Why Considered	Decision
LLaVA-1.6 (Mistral 7B)	Vision-LM	7B	Strong vision, open weights	✓ Primary Base
BioViL-T	Medical Vision	430M	Medical domain pretraining	✓ Encoder for embeddings
Qwen2-VL 7B	Vision-LM	7B	Strong multilingual (Urdu)	✓ Fallback / multilingual
GPT-4o Vision	Closed API	—	High accuracy	✗ No fine-tune access
InternVL2	Vision-LM	8B	SOTA on medical VQA	Evaluate in Phase 2

Table 6 — Base Model Candidates

LoRA Configuration

Parameter	Value	Rationale
LoRA Rank (r)	16	Balance between capacity and efficiency
LoRA Alpha (α)	32	Scaling factor = 2x rank, standard practice
Target Modules	q_proj, v_proj, k_proj, o_proj	Attention layers — highest impact
Dropout	0.05	Light regularization for medical domain
Learning Rate	2e-4	With cosine scheduler, warmup 100 steps
Batch Size	4 (grad accum 8)	Effective batch = 32
Epochs	3–5	Early stopping on val loss
Optimizer	AdamW 8-bit (bitsandbytes)	Memory efficient

Table 7 — LoRA Hyperparameters

QLoRA — Mobile/Edge Inference

After LoRA fine-tuning on the server, the model is quantized to 4-bit using bitsandbytes NF4 quantization. This reduces the 7B model from ~14GB to ~4GB, enabling inference on a single consumer GPU or cloud T4 instance — keeping API costs low for a Pakistan/UAE startup budget.

- Step 1: Load base model in 4-bit NF4 precision using BitsAndBytesConfig
- Step 2: Apply LoRA adapters on top of quantized base (PEFT library)
- Step 3: Fine-tune — only LoRA weights are updated, base stays frozen in 4-bit
- Step 4: Save merged model using merge_and_unload() for production deployment
- Step 5: Deploy on AWS g4dn.xlarge (T4 GPU) or RunPod for ~\$0.40/hour

Output — Condition Classification Schema

Condition Label	Severity	Action Triggered	Confidence Threshold
Healthy	None	Maintenance reminder only	> 0.85
Plaque / Tartar	Mild	Product suggest + brushing alarm	> 0.80
Early Cavity	Moderate	Product + dentist recommend (2 weeks)	> 0.78
Advanced Cavity	High	Dentist urgent referral (< 1 week)	> 0.75
Gingivitis	Moderate	Antibacterial product + dentist	> 0.78
Severe Gum Disease	Critical	Immediate dentist referral	> 0.72
Discoloration	Cosmetic	Whitening product suggestion	> 0.82
Unknown / Low Confidence	—	Request clearer photo + fallback	< threshold

Table 8 — Condition Classification Output Schema

05 — Hybrid RAG Pipeline

Dense + Sparse + Graph Retrieval

Simple vector search (dense RAG) is not enough for DantShaaf. A user's problem involves multiple dimensions — their diagnosed condition, severity, history, location, budget, and past product usage. Hybrid RAG combines three retrieval strategies to serve the most relevant products and dentists with full context awareness.

Three Retrieval Engines

Engine	Technology	What It Retrieves	Strength
Dense Retrieval	Qdrant + text-embedding-3-small	Semantically similar products, symptoms	Handles natural language queries
Sparse Retrieval	BM25 (Elasticsearch)	Exact keyword matches in product descriptions	Precise ingredient / brand name match
Graph Retrieval	Neo4j Cypher queries	Symptom→Treatment→Product →Dentist chains	Multi-hop reasoning, patient history

Table 9 — Three Retrieval Engines

Hybrid Fusion — How Results Are Combined

Results from all three engines are fused using Reciprocal Rank Fusion (RRF). Each engine returns a ranked list. RRF assigns a score of $1/(k + \text{rank})$ to each result and sums scores across engines. This is parameter-free, robust to score scale differences, and consistently outperforms single-engine retrieval by 15–25% on NDCG@10.



Knowledge Graph Schema (Neo4j)

The graph is the backbone of DantShaaf's intelligence. It connects patients, symptoms, treatments, products, and dentists into a traversable knowledge network. Every dentist upload creates new nodes. Every user session adds edges.

Node Type	Key Properties	Relationships
Patient	user_id, age, history[], severity_history	HAS_CONDITION, USED_PRODUCT, VISITED_DENTIST
Condition	label, severity, frequency	TREATED_BY, LINKED_TO
Treatment	name, type (product/procedure)	REQUIRES_PRODUCT, PERFORMED_BY
Product	name, brand, dentist_id, rating, price	SOLVES_CONDITION, UPLOADED_BY
Dentist	name, specialty[], rating, location_lat_lng	TREATS_CONDITION, LISTED_PRODUCT
Review	score, outcome_verified, date	ABOUT_DENTIST, ABOUT_PRODUCT

Table 10 — Neo4j Knowledge Graph Schema

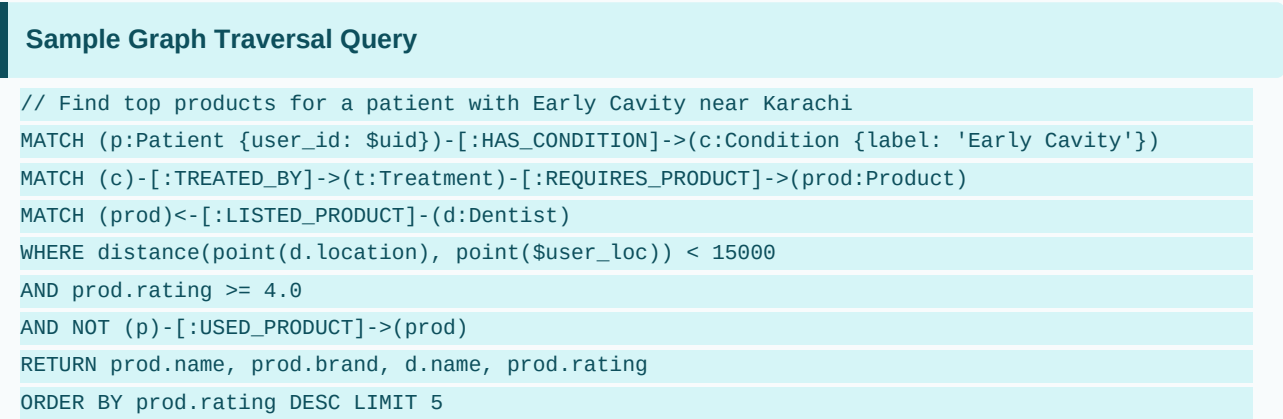


Figure 1 — Sample Cypher Query: Product recommendation for Early Cavity patient near Karachi

06 — AI Agent Design & MCP Tool Layer

The Autonomous Brain of DantShaaf

The DantShaaf Agent is built on Claude Sonnet 4 with a carefully designed system prompt that gives it a caring, Urdu-aware dental assistant persona. It is not a chatbot that responds — it is an agent that acts. All autonomous actions are executed through MCP (Model Context Protocol) tools.

MCP Tool Registry

Tool Name	Input	Output	Side Effect
analyze_teeth_image	base64 image, user_id	Condition label, severity, confidence	Logs to patient history DB
query_product_graph	condition, severity, user_id, location	Top-5 products with scores	None
find_nearest_dentist	condition, user_location, radius_km	Top-3 dentists with ratings	None
book_appointment	dentist_id, user_id, preferred_time	Confirmation, booking ID	Creates calendar event
set_alarm	user_id, message, datetime, recurrence	Alarm ID	Sets phone/WhatsApp reminder
log_progress	user_id, condition, date, notes	Progress entry ID	Updates progress timeline
send_whatsapp	user_phone, message_template, vars	Delivery status	Sends WhatsApp message
get_user_history	user_id	Full patient history JSON	None
escalate_to_dentist	user_id, severity, urgency	Escalation notice sent	Notifies registered dentist

Table 11 — Full MCP Tool Registry

Agent Decision Logic — Severity-Based Routing

Severity Level	Confidence	Primary Action	Secondary Action	Autonomous Follow-up
None (Healthy)	High	Maintenance tip	Whitening product (optional)	Monthly reminder
Mild	High	Product suggestion	Brushing alarm daily	Weekly progress check
Moderate	High	Product + Dentist recommend	2-week appointment reminder	Bi-weekly progress photo
High	High	Urgent dentist referral	Book appointment offer	3-day follow-up
Critical	Any	Immediate dentist alert	Emergency booking	Daily check-in
Any	Low (below threshold)	Re-request clearer photo	Suggest natural light tip	None until re-analysis

Table 12 — Severity-Based Agent Routing



Key Design Principle: The agent NEVER makes the user take extra steps. If a dentist should be booked, the agent offers to book it in the same message. If an alarm should be set, it is set without asking. Friction is the enemy.

07 — Dentist Portal & Product Graph

The B2B Revenue Engine

The Dentist Portal is a Next.js web application where registered dental professionals upload products, map them to conditions, and manage their clinic profile. Every upload feeds directly into the Neo4j knowledge graph, making it immediately available for RAG queries. This is the primary B2B revenue surface.

Portal Feature	What It Does	Tech Used
Product Upload Form	Name, description, condition tags, price, photos	Next.js + FastAPI
Condition Mapping	Link product to 1–5 dental conditions	Neo4j relationship creation
Analytics Dashboard	How many users were recommended this product, conversion rate	PostgreSQL + Chart.js
Clinic Profile	Location, specialties, availability, ratings display	Google Maps API
Patient Referral Feed	See patients who were referred by DantShaaf agent	Real-time PostgreSQL
Appointment Calendar	Manage bookings from DantShaaf users	Calendly API / Custom
Payout Dashboard	Track commission per referral and product recommendation	Stripe / JazzCash

Table 13 — Dentist Portal Feature Set

08 — Dentist Discovery & Nearby Recommendation

Matching Patients to the Right Specialist

When a patient's condition is Moderate, High, or Critical, the agent automatically queries for the best-matched dentist nearby. The matching algorithm combines three signals: geographic proximity, overall rating, and specialty-condition alignment from the knowledge graph.

Matching Algorithm — Three-Signal Scoring

Signal	Weight	Source	Formula
Specialty Match	50%	Neo4j graph — dentist specialties vs. condition	Binary match = 1.0, partial = 0.5
Rating Score	30%	DantShaaf reviews + Google Places merge	normalized_rating / 5.0
Proximity Score	20%	Haversine distance, user GPS vs dentist location	1 - (distance_km / max_radius_km)

Table 14 — Dentist Scoring Algorithm

Final Score = (0.50 × specialty_match) + (0.30 × rating_score) + (0.20 × proximity_score)

Data Sources for Ratings

- MVP Phase:** Google Places API — real ratings pulled at query time. No manual data entry needed. Dentists shown regardless of portal registration.
- Growth Phase:** DantShaaf in-app reviews from verified patients who used the app to visit the dentist. Weighted higher than Google (outcome-verified).
- Scale Phase:** AI-verified review system — NLP filters fake reviews, outcome tracking (did the condition improve post-visit?) adjusts ratings dynamically.

09 — Full Conversation Flow

End-to-End User Journey

Step	User Action	Agent Action	Tool Called	Outcome
1	Says: 'Mera dant dard kar raha hai'	Asks clarifying question about pain type	get_user_history	Context loaded
2	Describes thanda pain	Requests photo	—	Photo request sent
3	Uploads teeth photo	Analyzes image	analyze_teeth_image	Condition: Early Cavity, 0.84 confidence
4	Receives diagnosis	Queries products	query_product_graph	Top 3 products returned from RAG
5	Reads product suggestion	Queries nearby dentist	find_nearest_dentist	Top 2 dentists matched
6	Sees dentist options	Offers to book + sets alarm	set_alarm + book_appointment	Alarm set, booking offered
7	Confirms booking	Books appointment	book_appointment	Confirmation sent via WhatsApp
8	3 days later (automatic)	Follow-up message sent	send_whatsapp	Progress check initiated
9	Reports improvement	Logs progress	log_progress	Progress timeline updated
10	7 days later	Re-analysis request	analyze_teeth_image (new photo)	Progress delta computed

Table 15 — End-to-End Conversation Flow with Tool Calls

10 — Tech Stack & Infrastructure

Every Tool, Every Layer

Layer	Technology	Purpose	Cost Estimate
Agent LLM	Claude Sonnet 4 (Anthropic API)	Agent brain, tool orchestration, persona	\$0.003/1K tokens
Vision Fine-tune	LLaVA-1.6 + LoRA (PEFT library)	Teeth image analysis — local inference	One-time training cost
Quantization	bitsandbytes QLoRA 4-bit	Reduce model to 4GB for cheap inference	Included in PEFT
Dense Retrieval	Qdrant Cloud	Vector search for products/symptoms	Free tier → \$25/mo
Sparse Retrieval	Elasticsearch 8.x	BM25 keyword search	Self-hosted or \$16/mo
Graph Database	Neo4j AuraDB	Knowledge graph for RAG	Free tier → \$65/mo
Relational DB	PostgreSQL (Supabase)	User data, history, sessions	Free → \$25/mo
Inference Hosting	RunPod / AWS g4dn.xlarge	QLoRA model serving	~\$0.40/hr on demand
Backend API	FastAPI (Python)	All business logic, tool endpoints	Open source
Mobile App	React Native + Expo	iOS + Android from one codebase	Open source
Dentist Portal	Next.js 14	Web portal for dentists	Open source
Notifications	WhatsApp Business API (Meta)	Autonomous follow-ups	\$0.005–0.09/message
Location/Maps	Google Maps + Places API	Dentist discovery, geo-search	\$0.017/request
Payments	JazzCash / Stripe	Pakistan + international billing	2.9% + fixed fee
Auth	Supabase Auth / Clerk	User + dentist authentication	Free tier
MCP Server	Python FastMCP	Tool interface for Claude agent	Open source

Table 16 — Full Technology Stack

11 — Business Model & Revenue Streams

How DantShaaf Makes Money

Revenue Stream	Model	Pakistan Price	UAE Price	Notes
User Subscription	Freemium SaaS	Free / Rs.299/mo / Rs.799/mo	Free / AED 15/mo / AED 40/mo	Basic free, Pro = progress tracking + products
Dentist SaaS Portal	B2B Subscription	Rs.5,000–15,000/mo	AED 250–750/mo	Analytics, listing, priority RAG placement
Appointment Lead Fee	Per Referral	Rs.300–600/referral	AED 15–30/referral	Dentist pays per confirmed DantShaaf patient
Product Commission	Affiliate %	15–25% of product sale	15–25% of sale	When user buys via DantShaaf link
Premium Listing	Placement Ads	Rs.10,000–30,000/mo	AED 500–1500/mo	Top dentist/product placement in suggestions
Clinic Management Add-on	SaaS Upsell	Rs.8,000/mo	AED 400/mo	Full scheduling + patient record system
Dataset Licensing	Enterprise B2B	Custom pricing	Custom pricing	Anonymized dental data for research

Table 17 — Revenue Stream Matrix

12 — MVP Roadmap: Phase by Phase

From Zero to Launch in 12 Weeks

Phase 1 — Weeks 1–2: Foundation

- FastAPI backend setup with PostgreSQL + Qdrant
- Claude agent with basic dental persona and conversation flow
- Basic teeth photo analysis using Claude Vision (pre-fine-tune)
- User auth with Supabase

Phase 2 — Weeks 3–4: Fine-Tuning Pipeline

- Curate dental dataset from UFBA + Kaggle sources
- Run LoRA fine-tuning on LLaVA-1.6 using Hugging Face PEFT
- QLoRA quantization and deploy on RunPod
- Replace Claude Vision with fine-tuned model for diagnosis

Phase 3 — Weeks 5–6: Hybrid RAG

- Set up Neo4j AuraDB with initial condition/product/dentist graph
- Implement BM25 with Elasticsearch for sparse retrieval
- Build RRF fusion layer in FastAPI
- Manually seed 30–50 products and 20 dentists for testing

Phase 4 — Weeks 7–8: MCP Tools & Automation

- Build all 9 MCP tools (analyze, query, alarm, book, log, etc.)
- Connect WhatsApp Business API for follow-up messages
- Implement autonomous alarm setting via MCP
- Full end-to-end agent flow test

Phase 5 — Weeks 9–10: Dentist Portal

- Next.js dentist portal with product upload + condition mapping
- Google Places API integration for non-portal dentists
- Appointment booking flow (Calendly or custom)
- Analytics dashboard for dentists

Phase 6 — Weeks 11–12: Mobile App & Launch

- React Native mobile app (iOS + Android)
- Urdu language support in agent persona
- Beta launch with 10 dentist partners in Karachi/DHA
- Paid user acquisition via Meta ads

13 — Risk Analysis & Mitigations

Known Risks and How to Handle Them

Risk	Likelihood	Impact	Mitigation
Fine-tuned model gives wrong diagnosis	Medium	High	Confidence threshold — if <0.72, always recommend dentist; never act on low-confidence alone
User relies on AI instead of dentist	Medium	High	Mandatory disclaimer in every diagnosis; agent always says 'I recommend seeing a dentist for confirmation'
Dentist data not enough at launch	High	Medium	Google Places API covers all dentists even without portal registration; portal is additive
Model bias on Pakistani skin tones / photo quality	Medium	Medium	Augment training data with diverse lighting/angle samples; mobile UI guides user on photo quality
WhatsApp API policy violations	Low	High	Follow Meta's health messaging guidelines; users opt-in explicitly; all messages are user-initiated flows
Cold start — no product data in graph	High	Medium	Manually seed 50 products pre-launch; offer dentists 3-month free portal to build supply
Regulatory issues (medical advice)	Low	Very High	Position as 'awareness tool' not 'medical diagnosis'; all outputs include professional consultation disclaimer

Table 18 — Risk Register

14 — Appendix: Glossary

Technical Terms Explained

Term	Definition
LoRA	Low-Rank Adaptation. A fine-tuning technique that adds small trainable matrices to frozen model weights, reducing GPU memory and training time by 10–100x.
QLoRA	Quantized LoRA. Combines 4-bit quantization of the base model with LoRA adapters, enabling fine-tuning on consumer GPUs with minimal accuracy loss.
Hybrid RAG	Retrieval-Augmented Generation using multiple retrieval engines (dense vector search, sparse BM25, graph traversal) fused via RRF for superior results.
Graph RAG	RAG where the knowledge base is a graph database (Neo4j). Enables multi-hop reasoning — e.g. Symptom → Treatment → Product → Dentist chains.
MCP	Model Context Protocol. Anthropic's open standard for connecting AI agents to external tools and data sources via a standardized server interface.
RRF	Reciprocal Rank Fusion. A parameter-free method to combine ranked lists from multiple retrieval engines: score = sum of 1/(k+rank) across engines.
PEFT	Parameter-Efficient Fine-Tuning. A family of techniques (LoRA, prefix-tuning, adapters) that fine-tune large models by updating only a small subset of parameters.
NF4	Normal Float 4-bit quantization used in QLoRA. Stores model weights in 4 bits using a normally-distributed quantization grid, minimizing quantization error.
NDCG	Normalized Discounted Cumulative Gain. A ranking quality metric used to evaluate RAG retrieval — measures whether the most relevant results appear at the top.
Vision-LM	Vision-Language Model. A multimodal model (e.g. LLaVA, Qwen2-VL) that processes both images and text, used here for teeth photo analysis.